CSCE 581: Introduction to Trusted AI

Week 15 - Lectures 29: Final Session

PROF. BIPLAV SRIVASTAVA, AI INSTITUTE

30 APRIL, 2026

Carolinian Creed: "I will practice personal and academic integrity."

Credits: **Copyrights of all material reused acknowledged**

Organization of Lectures 30

- Main Message of Course
- Finals - Q3
- Concluding Section
  - Ask me anything

17	Mar 17 (Tu)	**Invited Guest** –	W10 Quiz 2 - start
18	Mar 19 (Th)	AI - Unstructured (Text): Processing and Representation	
19	Mar 24 (Tu)	AI - Unstructured (Text): Representation, Common NLP Tasks, Large Language Models (LLMs)	[Week 11] Quiz 2- end
20	Mar 26 (Th)	Natural Languages/ Language Models and their Impact on AI	
21	Mar 31 (Tu)	AI - Unstructured (Text): Analysis – Supervised ML – Trust Issues	[Week 12]Project – Sprint 2 - end
22	Apr 2 (Th)	AI - Unstructured (Text): Analysis – Supervised ML – Mitigation Methods	Project – Sprint 3 - start
23	Apr 7 (Tu)	AI - Unstructured (Text): Analysis – Rating and Debiasing Methods	[Week 13] Quiz 3 - end
24	Apr 9 (Th)	Explanation Methods Trust: AI Testing	
25	Apr 14 (Tu)	Trust: Human-AI Collaboration	[Week 14] Quiz 3 - end
26	Apr 16 (Th)	Emerging Standards and Laws Trust: Data Privacy - Trusted AI for the Real World	
27	Apr 21 (Tu)	Project presentation	[Week 15] Project – Sprint 3 - end
28	Apr 23 (Th)	Project presentation	Last day of class
	Apr 28 (Tu)		Reading Day
29	April 30 (Th)	12:30pm – Final exam/ Overview	[Week 16]Optional, information shared

#1 NEWS – Talkies (LLM trained on 1930 era data)

Link - https://www.marktechpost.com/2026/04/27/meet-talkie-1930-a-13b-open-weight-llm-trained-on-pre-1931-english-text-for-historical-reasoning-and-generalization-research/

- **Context**
  - 13-billion parameter open-weight language model trained exclusively on pre-1931 English text
  - **December 31, 1930** — date when works enter the public domain in the United States, making pre-1931 text legally usable for training.
- **Findings**
  - Events after 1930 — talkie's knowledge cutoff — are consistently more surprising to the model, with the effect most pronounced for 1950s and 1960s events, followed by a plateau.
  - This creates a principled setup for studying how forecasting ability scales with model size and how performance decays over longer temporal horizons.
  - Allows clean experimental setting to test how well an LM can generalize beyond its pre-training data. For example, the team tested whether talkie could learn Python — a language that didn't exist in 1930 — by providing a few in-context demonstration examples. Using the HumanEval benchmark, they found that while vintage models dramatically underperform web-trained models, they are "slowly but steadily improving at this task with scale."
- **Insight**
  - New possibilities for research if we have data and know how to use.
  - Another example: Lighthouse (Civil Rights Explorer at USC)

High Level Plan (Original)

CSCE 581 –

- Week 1: Introduction
- Week 2: Background: AI - Common Methods
- Week 3: The Trust Problem
- **Week 4: Machine Learning (Structured data) - Classification**
- **Week 5: Machine Learning (Structured data) - Classification – Trust Issues**
- **Week 6: Machine Learning (Structured data) – Classification – Mitigation Methods**
- **Week 7: Machine Learning (Structured data) – Classification – Explanation Methods**
- Week 8: Machine Learning (Text data) – Classification, **Large Language Models**
- **Week 9: Machine Learning (Text data) - Classification – Trust Issues**
- **Week 10: Machine Learning (Text data) – Classification – Mitigation Methods**
- **Week 11: Machine Learning (Text data) – Classification – Explanation Methods**
- Week 12: Emerging Standards and Laws
- Week 13: Project presentations
- Week 14: Project presentations, Conclusion

AI/ ML topics and with a
focus on fairness, explanation,
Data privacy, reliability

Student Assessment

A = [920-1000]
B+ = [870-919]
B = [820-869]
C+ = [770-819]
C = [720-769]
D+ = [670-719]
D = [600-669]
F = [0-599]

Tests	Undergrad	Grad
Course Project – report, in-class presentation	600	600
Quiz – best of 2 from 3	200	200
Final Exam	200	100
Additional Final Exam – Paper summary, in-class presentation		100
Total	1000 points	1000 points

Grades and Attendance Policy

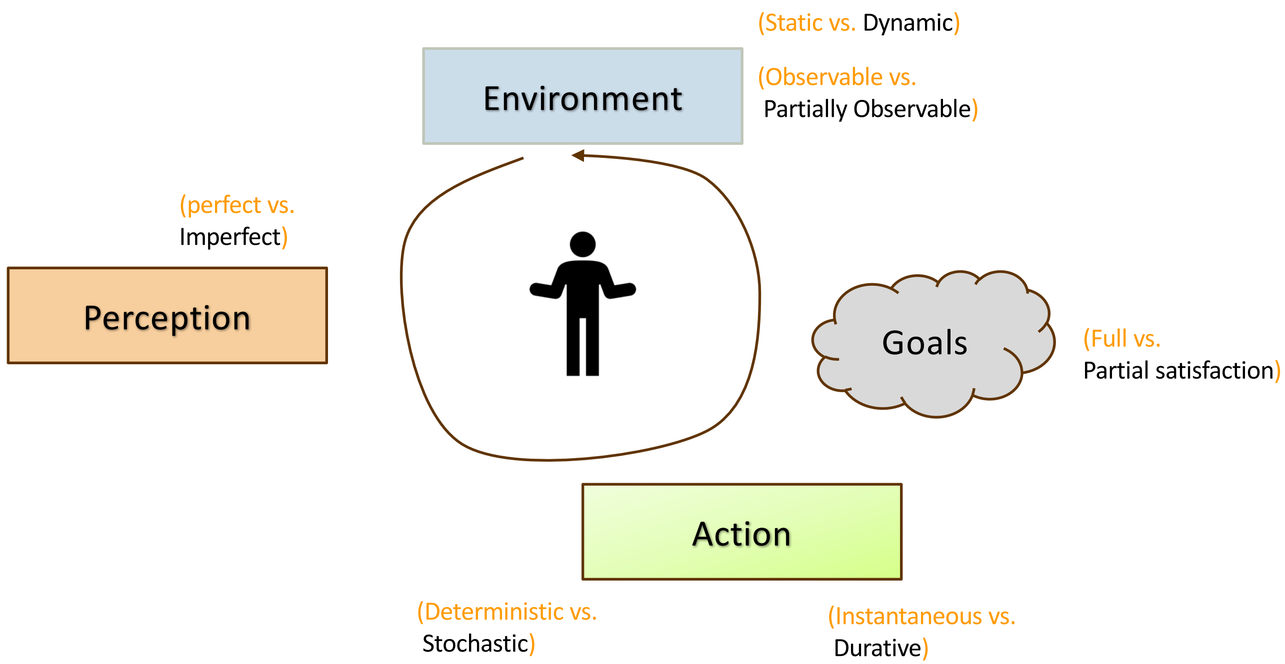
As an in-person class, I will impose a **common-sense**, **necessary – but not sufficient**, attendance policy for those taking the class for grades.

will be out of 14 classes:

1. 80% or more classes attending: **A, B+ or B**
2. 50-79% classes: **B+, B, C+, C, D+ or D**
3. 49% or below: **F**

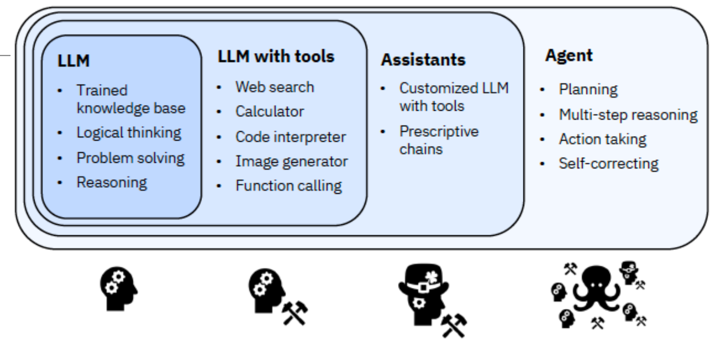
Main Message From Course

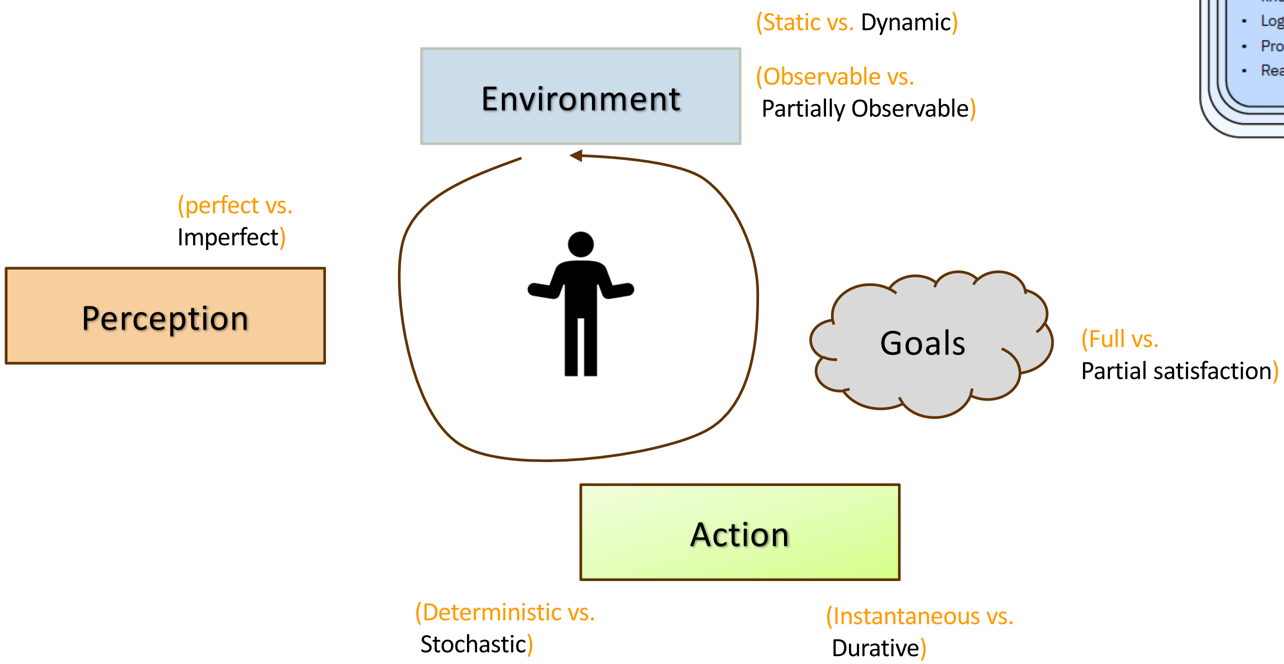
Intelligent Agent Model



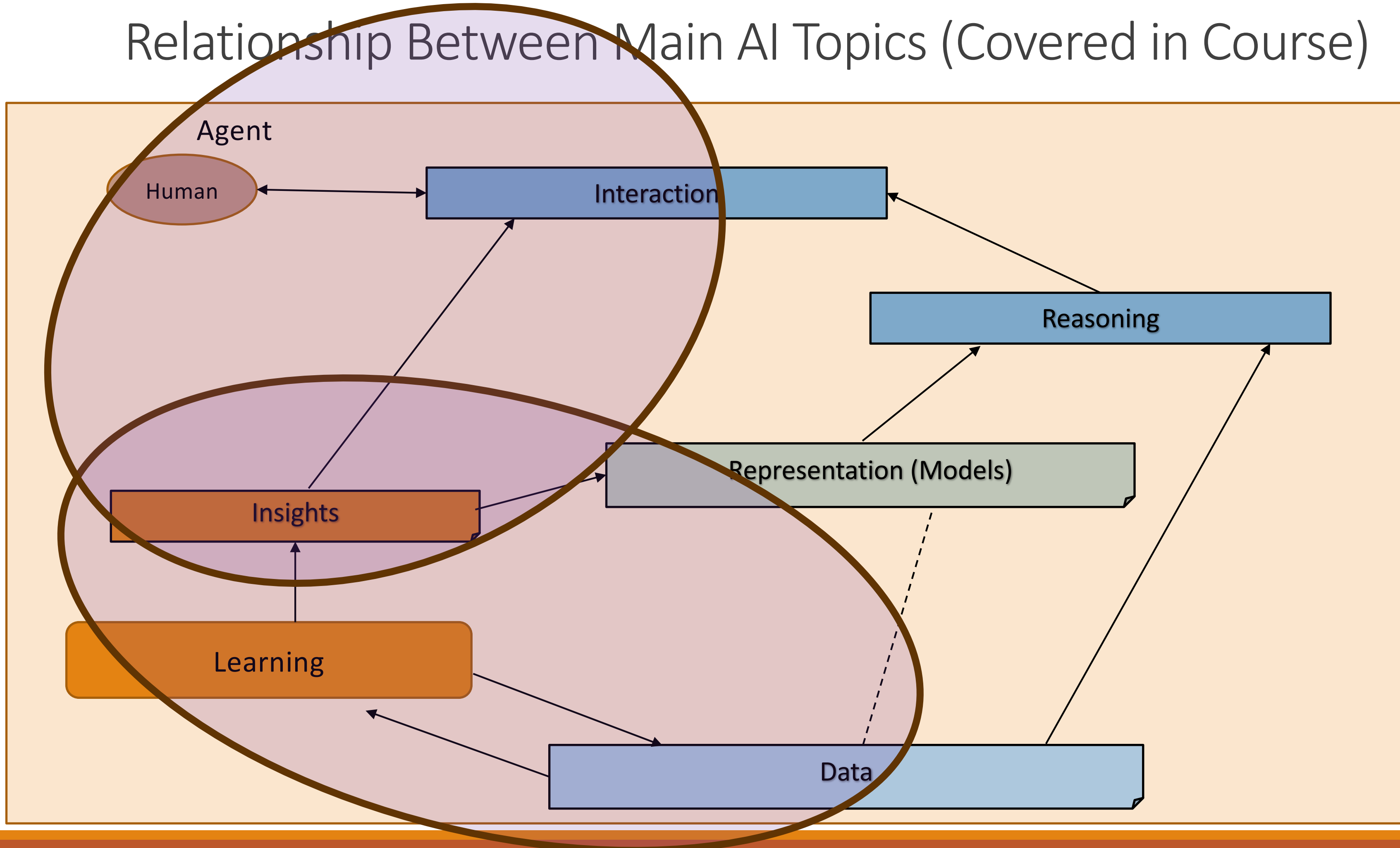
Intelligent Agent Model vis-a-vis Agentic AI

Credit: K. Varshney's talk - **Individual and Collective Human Agency in the Face of 'AI'**, 2026

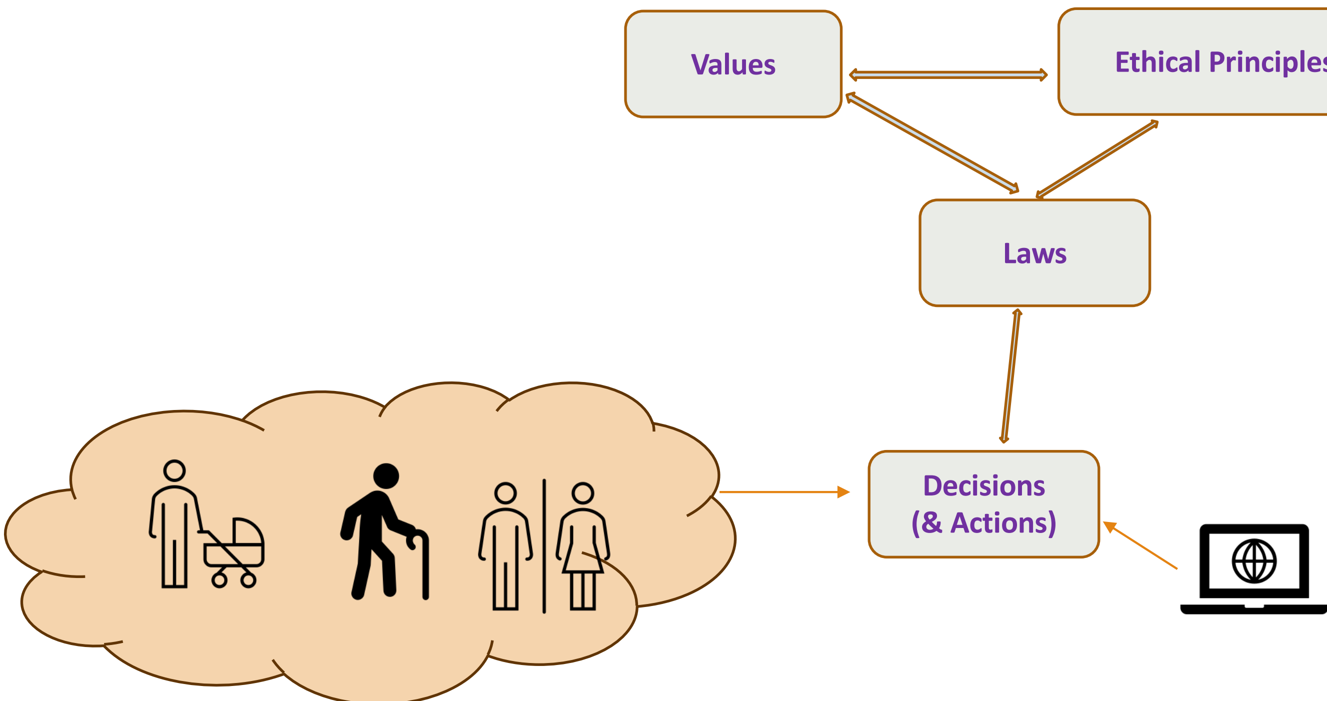




Relationship Between Main AI Topics (Covered in Course)



Values, Ethics, Law, Decisions (Actions) with Trust



Trust: *how can,*
humans or machines (AI) or in collaboration together,
take decisions and actions following the same ***values***, ***ethical*** *principles and* ***laws ?***
Assumption: *they should follow same values, principles and laws.*

Main AI Ethics Issues

- DATA GOVERNANCE AND PRIVACY
- FAIRNESS AND INCLUSION
- HUMAN AND MORAL AGENCY
- VALUE ALIGNMENT
- ACCOUNTABILITY
- TRANSPARENCY AND EXPLAINABILITY
- TECHNOLOGY MISUSE

Credits:
Tutorial on Trusting AI by Testing and Rating Third Party Offerings at IJCAI 2020, Biplav Srivastava, Francesca Rossi, Jan 2021

Finals, Q3

Case to Consider

Choice 1

- Role: Chief AI Officer @University
- Candidates: Any faculty who has AI in their biodata and whose name comes up in sc.edu page

Choice 2

- Role: Coach for University Team in basketball
- Candidates: Any coach in university team across the US

Ask-Me-Anything (AMA)